P3 October 2025 + ESAT — Jargon & Formula Recall Pack

Harry → Phuc · Tutorial of 11 July 2026 · Edexcel IAL Pure P3 (WMA13/01, 21 Oct 2025)

Compiled 14 July 2026

Contents

0. How to read this pack

This is a **recall pack**: dense, cross-tabulated definitions and formula/method tables for the 11 July 2026 tutorial, meant for fast pre-exam review and for a native-Notion table export. Every statement is grounded in one of three source tiers, and each is tagged so you can tell tutor voice from official text:

- **[T] Tutor-taught** — what Harry actually said or wrote in the Loom recording (spoken transcript `sources/loom_transcript.txt`, corroborated by the four-segment video ledger `analysis/gemini_video_evidence.md`). This is the primary evidence and is never overwritten.

- **[Q] Official-QP** — verbatim from the question paper *Pearson Edexcel IAL Pure Mathematics P3, WMA13/01, Tue 21 Oct 2025* (`sources/WMA13_Oct2025_QP.txt`). All question numbers, forms and command words are quoted from this paper.
- **[S] Official / spec-derived reference** — from the *Edexcel IAL Mathematics 2018 specification* and its formula-book table. Used **only** to expand the formal statement behind a method (e.g. the spec's own wording of "location of roots"). These expansions are always labelled **[S]** and are **not** put into Harry's mouth.

Where the evidence pack is internally inconsistent, the conflict is flagged, not smoothed over. The four live conflicts (ESAT option labels; Q4(a) coefficients; the Q5 denominator; the "459 vs 959" maximum) and the one **status** conflict (Q2(b) completion) are resolved in §7 exactly as in the parent guide and `evidence_coverage.md`. Three status rules are load-bearing and repeated throughout:

- **Q2(b)** = *completed with prompting — transcript/ledger status conflict* (transcript: Harry confirms "we're there"; video ledger calls it partial).
- **Q2(c)** = *skipped / deferred* — explicitly left ("we'll leave that one"), never started. No worked iteration is invented here.
- **Q1** = *discussed only*. No tutor working exists, so none is reconstructed.

---

1. Jargon glossary

Definitions are 30–80 words. The final column is **Phuc trap** only where the recording shows Phuc actually making that error; otherwise it is a **Common trap** (a standard method pitfall Harry warned about in general, or an examiner convention). Tier tags: [T] tutor, [Q] paper, [S] spec.

Term	Student-friendly meaning	Exact mathematical meaning	Symbols / conditions	What it is NOT	Lesson example	Phuc / common trap
Surface area [T]	The total area of all the outside faces of a solid — the amount of "skin".	Sum of the areas of every bounding surface of a 3-D solid; measured in units². For a closed solid it includes *every* face.	Sphere $4\pi R^2$; closed cylinder $2\pi r^2 + 2\pi rh$; units are (length)².	Not the *volume* (space inside) and not a single face — a closed cylinder has **two** flat ends, not one.	ESAT warm-up: sphere skin $= 10\times$ one cylinder's skin, with $h = 4R$.	**Phuc trap:** offered $\pi r^2 h$ (the *volume*) as the cylinder's surface area; Harry: "this is the volume… we're just interested in surface area."
Volume of a prism [T]	Cross-section area pushed straight through the height.	For any prism (constant cross-section), $V = (\text{cross-section area}) \times (\text{perpendicular height})$. A cylinder is a prism with a circular cross-section, so $V = \pi r^2 h$.	$V = \pi r^2 h$; needs a *constant* cross-section parallel to the ends.	Not a surface area; $\pi r^2 h$ is space filled, not skin.	Harry derived it to explain *why* $\pi r^2 h$ is volume, then set it aside — the ESAT question wanted surface area only.	**Common trap:** quoting the volume formula when the question asks for surface area (and vice-versa).
Circumference [T]	The distance once around the edge of a circle.	The perimeter of a circle, length $2\pi r$; equivalently arc length $r\theta$ with the full angle $\theta = 2\pi$.	$C = 2\pi r$ (= $r\theta$, $\theta = 2\pi$).	Not the **diameter** ($2r$), and not πr.	The unrolled curved side of a cylinder is a rectangle of width = circumference $2\pi r$.	**Phuc trap:** called the rectangle's long side the "diameter"; Harry: "No, it's not the diameter, it's the circumference."

Term	Student-friendly meaning	Exact mathematical meaning	Symbols / conditions	What it is NOT	Lesson example	Phuc / common trap
Net / unrolling [T]	Cut a solid open and lay it flat to see the real areas.	The 2-D flat pattern whose regions have the same areas as the curved surfaces of the solid. Unrolling a cylinder's curved side gives a rectangle $2\pi r$ by h.	Curved side area $= 2\pi r \times h = 2\pi rh$.	Not a change of size — areas are preserved, only re-shaped.	"Imagine wrapping paper… you cut it and unfurl it — what shape? A rectangle." → curved area $2\pi rh$.	**Common trap:** guessing the curved area instead of deriving it from the $2\pi r \times h$ rectangle.
Area of a circle [T]	The space inside the round boundary.	$A = \pi r^2$, r the radius.	$A = \pi r^2$.	Not πr and not the circumference $2\pi r$.	Two of these are the flat ends of the cylinder: $2 \times \pi r^2$.	**Phuc trap:** wrote "π multiplied by r" (πr); Harry: "Pi r *squared*… don't forget the square."
Sphere surface area [T][S]	The skin of a ball.	$A = 4\pi R^2$ for a sphere of radius R — "the total area around the outside of the 3-D shape."	$A = 4\pi R^2$; R the sphere radius (capital in the ESAT question).	Not $4\pi R$ (missing square) and not the volume $\frac{4}{3}\pi R^3$.	Left side of the ESAT word-equation: $4\pi R^2 = 10(2\pi r^2 + 2\pi rh)$.	**Phuc trap:** first wrote $4\pi r$ / "I forget this one"; Harry: "don't forget the square."
Intermediate Value Theorem (sign change + continuity) [T][S]	If a smooth graph goes from below the axis to above it, it must cross the axis in between.	[S] Spec 6.1: locate a root of $f(x) = 0$ by a **change of sign** of f across an interval **in which f is continuous**. Both conditions are required.	$f(a)f(b) < 0$ **and** f continuous on $[a, b] \Rightarrow$ a root $\alpha \in (a, b)$.	Not sign change alone — without stated continuity a sign change proves nothing (think of a jump).	Q2(a): evaluate $f(1), f(2)$, show opposite signs, state continuity on $[1, 2]$, conclude a root.	**Phuc trap:** stopped at "sign change" and forgot continuity. Harry: "the bit everyone forgets is continuity… otherwise you'll only score one mark."
Root / α [Q]	An x-value where the function equals zero (a graph crossing).	A solution of $f(x) = 0$; α denotes the specific root being located/approximated.	$f(\alpha) = 0$; here $\alpha \in [1, 2]$.	Not a turning point and not a y-value; it is the x where $f = 0$.	Q2 asks to *show* $\alpha \in [1, 2]$ (a), rearrange (b), then iterate to α (c).	**Common trap:** substituting values "near" the endpoints (1.001, 1.999) instead of the endpoints themselves — Harry: "just use the endpoints."
Iteration / recurrence [Q][S]	Feed each answer back in to get closer to the root, step by step.	[S] Spec 6.2: approximate solution of $f(x) = 0$ using a recurrence $x_{n+1} = f(x_n)$ from a start value x_1.	$x_{n+1} = \sqrt[3]{\dfrac{e^{x_n} + 4x_n^2}{2x_n + 3}}$, $x_1 = 1$ (Q2c).	Not an "arithmetic series"; terms are not evenly spaced, they converge.	**Q2(c) — skipped/deferred:** "we'll leave that one." No iterate values are computed in this pack.	**Common trap:** rounding intermediate iterates; quote x_3 and α each to 4 d.p. from unrounded values (deferred here).
"Show that" (command word) [T][Q]	Prove you reach the given result — the answer is already printed for you.	An instruction to demonstrate a stated result with full working; you must *arrive at exactly the printed expression* and stop there.	Stop at the shown line; do not solve further unless a later part asks.	Not an invitation to keep going (e.g. Q5(c) shows $e^{0.2T} = A$ — don't solve for T).	Q2(b), Q5(c) are "show that" parts; Q5(d) then says "hence find".	**Phuc trap:** on Q5(c) pushed on to solve for T; Harry: "we've done a bit more than we needed… go back."
Log-linear graph / linearisation [T][S]	Plot the log of a quantity so an exponential law becomes a straight line.	[S] Spec 3.3: for $y = kb^t$, plotting $\log_{10} y$ against t gives a line with intercept $\log_{10} k$ and gradient $\log_{10} b$.	Axis convention "Y against X" = Y vertical, X horizontal. Here $\log_{10} V$ (vertical) vs t (horizontal).	The vertical axis is **not** V and not y — it is $\log_{10} V$.	Q3(a): line through $(0, 2), (5, 2.25)$ gives $\log_{10} V = 0.05t + 2$.	**Phuc trap:** tried to use V (or y) as the vertical variable; Harry: "It's not V. Be careful… look at our axes."

Term	Student-friendly meaning	Exact mathematical meaning	Symbols / conditions	What it is NOT	Lesson example	Phuc / common trap
Index / log laws (undoing a log) [T]	Swap between "log of something" and "10 to a power" to free the variable.	$\log_{10} V = p \Leftrightarrow V = 10^p$; and $10^{p+q} = 10^p 10^q$, $(10^k)^t = 10^{kt}$.	$V = 10^{0.05t+2} = 10^2(10^{0.05})^t$.	Not "take log of both sides again"; you raise 10 to the power to *remove* the log.	Q3(b): $\log_{10} V = 0.05t + 2 \Rightarrow V = 100(10^{0.05})^t$, so $a = 100$, $b = 10^{0.05}$.	**Common trap:** leaving $10^{0.05t}$ un-restructured instead of writing it as $(10^{0.05})^t$ to match ab^t.
Exact value [T][Q]	Leave it as a surd/fraction/π — no rounded decimals.	A value written with no rounding: surds, fractions, π, e, ln kept symbolic.	e.g. $R = 2\sqrt{3}$, $\alpha = \frac{\pi}{3}$, $a = 100$, $b = 10^{0.05}$.	Not a decimal approximation (e.g. $R = 3.46$, $\alpha = 1.05$ are *not* exact).	Q4(a) demands exact R, α; Q3(b) demands exact a (but b to 3 s.f.).	**Phuc trap:** Harry: "exact means… no decimals." Preferring a decimal route is why the square-and-add method is safer (see §5).
Rate of change [T]	How fast something grows per unit time — a derivative, not a step.	The instantaneous rate is the derivative $\dfrac{d(\text{quantity})}{dt}$; a stated "rate of increase" is a positive derivative value.	Q3(c): $\dfrac{dV}{dt} = +50$ (increase $\Rightarrow +$, not -50).	Not an "arithmetic series" and not a discrete change; it is continuous.	"The rate of increase was £50 per year" $\Rightarrow \frac{dV}{dt} = 50$.	**Phuc trap:** guessed "arithmetic series / something linear"; Harry: "whenever you see *rate*… think $\frac{d\cdot}{dt}$."
Power function a^{kx} and its derivative [T][S]	Differentiating a *number* to a power brings down a ln of that number (and the chain factor k).	[S] Formula book: $\dfrac{d}{dx}a^x = a^x \ln a$. Chain rule extends it: $\dfrac{d}{dx}a^{kx} = a^{kx}\, k \ln a$.	$a > 0$ constant base; k constant. Factor is $k \ln a$.	The extra factor is $k \ln a$ — **not** $\ln(kx)$ and not $\ln k$.	Drills: $\frac{d}{dx}3^{6x} = 3^{6x} \cdot 6 \ln 3$; $\frac{d}{dt}1.12^t = 1.12^t \ln 1.12$.	**Phuc trap:** for 3^{6x} wrote "… $\ln(6x)$"; Harry: "Not natural log $6x$." Also: differentiate the *exact* base $10^{0.05}$, not the rounded 1.12.
$R\sin(2x - \alpha)$ **(harmonic / auxiliary-angle form)** [T][Q][S]	Rewrite "$a \sin -b \cos$" as one shifted sine wave.	[S] Spec 2.3: express $a \sin\theta - b\cos\theta$ as $R\sin(\theta - \alpha)$. Expand: $R\sin(2x-\alpha) = (R\cos\alpha)\sin 2x - (R\sin\alpha)\cos 2x$.	$R > 0$; here $0 < \alpha < \frac{\pi}{2}$. Match: $R\cos\alpha = \sqrt{3}$, $R\sin\alpha = 3$.	Not a formula to memorise raw — Harry: "look it up in the P3 formula book," mind the minus-sign version.	Q4(a): $\sqrt{3}\sin 2x - 3\cos 2x = 2\sqrt{3}\sin(2x - \frac{\pi}{3})$.	**Common trap:** forcing yourself to memorise the identity instead of reading it from the formula book with the correct $\pm$ sign.
Square-and-add (for exact R) [T]	To get R exactly, square both matched equations and add — the trig cancels.	From $R\cos\alpha = a$, $R\sin\alpha = b$: $(R\cos\alpha)^2 + (R\sin\alpha)^2 = R^2(\cos^2\alpha + \sin^2\alpha) = R^2 = a^2 + b^2$, so $R = \sqrt{a^2+b^2}$; and $\tan\alpha = b/a$.	Uses $\cos^2\alpha + \sin^2\alpha = 1$; gives R, α exactly.	Not "sub a decimal α back to get R" — that risks rounding error.	Q4(a): $(\sqrt{3})^2 + 3^2 = 12 \Rightarrow R = 2\sqrt{3}$; $\tan\alpha = 3/\sqrt{3} = \sqrt{3} \Rightarrow \alpha = \frac{\pi}{3}$.	**Phuc trap:** preferred substituting a decimal α to find R; Harry: "your method could be quite dangerous… square and add."
Principal value vs general solution [T]	arcsin gives one angle, but a sine equation has many solutions — check you have the *smallest*.	For $\sin\theta = 1$ the principal $\theta = \frac{\pi}{2}$; general solution $\theta = \frac{\pi}{2} + 2n\pi$. Also check the $\pi - \theta$ partner before rearranging.	$\theta = 6x - \frac{\pi}{3}$; smallest $x > 0$ wanted (Q4b-ii).	Not "the calculator value is automatically the smallest x" — you must justify it.	Q4(b)(ii): $6x - \frac{\pi}{3} = \frac{\pi}{2} \Rightarrow x = \frac{5\pi}{36}$ (next is $\frac{17\pi}{36}$).	**Common trap:** not verifying it is the smallest root. Here $\pi - \frac{\pi}{2} = \frac{\pi}{2}$ coincides, so one value — Harry: "in other questions you'd need the others."

Term	Student-friendly meaning	Exact mathematical meaning	Symbols / conditions	What it is NOT	Lesson example	Phuc / common trap
Minimise a fraction (optimisation) [T]	A fixed number over a wave is smallest when the bottom is biggest.	For $g = \dfrac{c}{D(x)}$ with $c > 0$ fixed, g is minimised when the denominator $D(x)$ is maximised; here D maximal when its sine term $= 1$.	min $g = \dfrac{18}{2\sqrt{3}(1) + 4\sqrt{3}} = \dfrac{18}{6\sqrt{3}} = \sqrt{3}$.	Not a calculus turning-point hunt — reason from the sine's max, no differentiation needed.	Q4(b)(i): min $g = \sqrt{3}$ (after rationalising $\frac{3}{\sqrt{3}}$).	**Common trap:** leaving $\frac{3}{\sqrt{3}}$ un-rationalised; multiply top and bottom by $\sqrt{3}$ to get $\sqrt{3}$.
Quotient rule [T][S]	Differentiating "top over bottom" — with a **minus** sign in the numerator.	[S] Formula book (uses f, g): $\left(\dfrac{u}{v}\right)' = \dfrac{v\,u' - u\,v'}{v^2}$. Order matters because of the minus.	$u = 4000e^{0.1t}$, $v = 19 + e^{0.2t}$; $u' = 400e^{0.1t}$, $v' = 0.2e^{0.2t}$.	Not the product rule on uv^{-1} (Harry advised against it here); and the sign is **minus**, not plus.	Q5(b): $\dfrac{dN}{dt} = \dfrac{7600e^{0.1t} - 400e^{0.3t}}{(19 + e^{0.2t})^2}$.	**Phuc trap:** dropped the minus sign (and floated rewriting as uv^{-1}); Harry: "a minus sign on the top — be careful… it's in the formula book."
Maximum / stationary point [T]	The top of the curve, where the gradient is momentarily flat (zero).	A local maximum of $N(t)$ satisfies $\dfrac{dN}{dt} = 0$. With a positive denominator squared, only the numerator can be zero.	At $t = T$: numerator $= 0$; $e^{0.1T} > 0$ always, so solve the bracket.	Not "denominator $= 0$" — $(19 + e^{0.2T})^2$ is never zero.	Q5(c): $7600 - 400e^{0.2T} = 0 \Rightarrow e^{0.2T} = 19$, so $A = 19$.	**Common trap:** trying to zero the whole fraction or the denominator; only the numerator's bracket gives the stationary point.
Rounding discipline (stored value, s.f., nearest whole) [T][Q]	Keep full precision until the very last line, then round as the question tells you.	Carry exact/stored values through; round *only* the final answer to the stated precision (3 s.f., or nearest whole number).	Q3 b: 3 s.f.; Q3(c),(d): nearest integer; Q5(d): nearest whole number.	Not rounding T then substituting — that corrupts the final answer.	Q5(d): use stored $T = \ln 19/0.2$; $N = \dfrac{4000\sqrt{19}}{38} = 458.83\ldots \to 459$.	**Phuc trap:** rounded T before substituting back; Harry: "we only round at the very final answer… use the stored value."
Formula book [T][S]	The official booklet of formulae you're given in the exam — look things up instead of memorising.	The Edexcel IAL formula book; its P3 section holds the compound-angle identities and the quotient rule (written with f, g).	Radians assumed; use $\tan^{-1}$ (not $\sin^{-1}$) for α.	Not a substitute for method fluency — you still need to know *which* formula and how to use it.	Harry pasted the R sin identity and pointed to the quotient rule "hiding" in the P3 section.	**Phuc trap:** calculator slips — misread own "3" as "1", used $\sin^{-1}$ for α, and had the calculator in the wrong mode.

---

2. ESAT surface-area & modelling table

Problem (as read on the board) [T]: the surface area of a solid sphere of radius R equals the total surface area of **10** solid **closed** cylinders of radius r, height h, with $h = 4R$. Find R in terms of r. The board's multiple-choice option labels were **not reliably captured** (see §7) — match the value to whichever lettered option appears on the day.

Formula / step	Conditions & meaning	Trigger (when to use)	Trap	Src
Sphere surface area $= 4\pi R^2$	Whole outside skin of a ball, radius R.	Any “surface area of a sphere”.	Keep the **square**; not $4\pi R$, not the volume.	[T][S]
Circle area $= \pi r^2$ (×2 for two ends)	Two flat circular ends of a *closed* cylinder.	Closed-cylinder surface area.	Not πr; a closed cylinder has **two** ends, not one.	[T]
Curved side $= 2\pi r \times h = 2\pi rh$	Unroll the curved face to a rectangle $2\pi r$ (circumference) by h.	Any cylinder’s curved surface.	Long side is circumference $2\pi r$, **not** the diameter.	[T]
Closed cylinder S.A. $= 2\pi r^2 + 2\pi rh$	Two ends + curved side.	“Surface area of a (closed) cylinder.”	Not $\pi r^2 h$ (that is the **volume**).	[T]
Word equation $4\pi R^2 = 10(2\pi r^2 + 2\pi rh)$	Sphere skin = 10 × one cylinder’s skin.	Translate the sentence before algebra.	Using one end ($\pi r^2 + 2\pi rh$) undercounts — needs both ends.	[T]
Sub $h = 4R$, ÷ 4π $\Rightarrow R^2 - 20rR - 5r^2 = 0$	Substitute the given link, simplify to a quadratic **in** R.	Once one unknown is expressed via another.	Treat as quadratic in R (with r constant), coefficients $a{=}1, b{=}-20r, c{=}-5r^2$.	[T]
$R = \dfrac{20r \pm \sqrt{400r^2+20r^2}}{2} = r(10 \pm \sqrt{105})$	Quadratic formula; $\sqrt{420} = 2\sqrt{105}$.	Solving the quadratic.	Simplify the surd fully; keep it exact.	[T]
$\boxed{R = r(10+\sqrt{105})}$	Radius must be positive $\Rightarrow$ take the $+$ root.	Final ESAT answer.	Rejecting the wrong root: $10 - \sqrt{105} < 0$ is invalid.	[T]

[T] Transferable point Harry stressed: “I think we need to know some key formulae.” The whole ESAT question collapses once the sphere and closed-cylinder formulae are known — his fix is flashcards + the cheat-sheet he sent.

---

3. IVT & iteration table (Q2)

From the paper [Q]: $f(x) = \dfrac{2x^2+3x-4}{e^x} - \dfrac{1}{x^2}$, $x \in \mathbb{R}, x \neq 0$. (a) show $f(x) = 0$ has a root $\alpha \in [1,2]$; (b) show it can be written $x = \sqrt[3]{\dfrac{e^x + 4x^2}{2x+3}}$; (c) iterate with $x_1 = 1$ to find x_3, α to 4 d.p. **[7 marks]**

Method / line	Conditions & meaning	Trigger	Trap	Src
(a) Evaluate $f(1)$ and $f(2)$ at the **endpoints**	Show one value $+$ve, one $-$ve = a sign change.	“Show a root lies in $[a,b]$.”	Use the actual endpoints, **not** 1.001/1.999.	[T]
(a) State f is **continuous** on $[1,2]$	Second required condition of location-of-roots (spec 6.1).	Immediately after the sign change.	Omitting continuity caps you at **1 mark** — “the bit everyone forgets.”	[T][S]
(a) Conclude: root $\alpha \in [1,2]$	Sign change + continuity $\Rightarrow$ a root exists.	To close the argument.	Not concluding explicitly.	[T][S]
(b) Set $f(x) = 0$; move $\frac{1}{x^2}$ across; $\times e^x x^2$	Clear denominators toward the target form.	“Show it can be written as…” — work *toward* the printed goal.	Don’t compute $f(0)$; the equation is $f(x) = 0$ for general x.	[T][Q]

Method / line	Conditions & meaning	Trigger	Trap	Src
(b) $2x^4+3x^3=e^x+4x^2$; factor $x^3(2x+3)$	The $2x+3$ in the target denominator signals factoring x^3 out.	Spotting $2x+3$ in the goal.	See the denominator as the clue: "where's the $2x+3$ coming from?"	[T]
(b) $x^3=\dfrac{e^x+4x^2}{2x+3}$; cube-root both sides	A cube root in the answer ⇒ cube-root both sides.	Final step to the printed form.	Cube-rooting *before* forming the single fraction leaves the division undone.	[T]
(b) STATUS	**Completed with prompting — transcript/ledger conflict.** Transcript: Harry "well done, and then we're there, aren't we?"; video ledger marks it *partial*.	—	Do not downgrade to "skipped"; the *skipped* item is (c).	[T]
(c) $x_{n+1}=\sqrt[3]{\dfrac{e^{x_n}+4x_n^2}{2x_n+3}}$, $x_1=1$	Recurrence approximating α (spec 6.2).	"Using the iteration formula…"	**Skipped / deferred** — "we'll leave that one." No iterates computed here.	[T][Q][S]
(c) *(when you do it)* keep calc mode; don't round iterates	Quote x_3 and α each to 4 d.p.	Homework: prove it and send to Harry.	Rounding mid-iteration; this is a *transcript* invitation, not a study-log task (see §7).	[T]

4. Exponential / log models & differentiation table (Q3)

From the paper [Q]: graph of $\log_{10} V$ against t is a straight line through $(0,2)$ and $(5,2.25)$. (a) find $\log_{10} V=mt+c$; (b) write $V=ab^t$ (exact a, b to 3 s.f.); (c) when $t=T$ the rate of increase is £50/yr — find T (nearest integer). *"Solutions relying entirely on calculator technology are not acceptable."* **[9 marks]**

Method / line	Conditions & meaning	Trigger	Trap	Src
Axes: $\log_{10} V$ vertical, t horizontal	"Y against X" ⇒ Y vertical.	Reading any "log against t" graph.	Vertical variable is $\log_{10} V$, **not** V or y.	[T]
(a) $m=\dfrac{2.25-2}{5-0}=0.05$, $c=2$	Gradient & intercept of the line.	Straight-line log graph.	Read c off the $(0,2)$ intercept.	[T]
(a) $\boxed{\log_{10} V=0.05t+2}$	Line equation in the required form.	Answer to (a).	Write $\log_{10} V$, never V, on the left.	[T][Q]
(b) $V=10^{0.05t+2}=100\,(10^{0.05})^t$	Raise 10 to each side; split index $10^{p+q}=10^p10^q$.	Converting a log line to ab^t.	Restructure $10^{0.05t}$ as $(10^{0.05})^t$ to expose b.	[T]
(b) $a=100$ (exact), $b=10^{0.05}=1.12$ (3 s.f.)	a exact, b rounded per the command.	Answer to (b).	Keep exact $10^{0.05}$ for part (c); don't differentiate the rounded 1.12.	[T][Q]
(c) "rate of increase" ⇒ $\dfrac{dV}{dt}=+50$	Rate = derivative; increase ⇒ positive.	The word "rate".	Not an arithmetic series; $+50$ not -50.	[T]

Method / line	Conditions & meaning	Trigger	Trap	Src
(c) $\dfrac{d}{dx}a^{kx} = a^{kx}k \ln a$	[S] $\frac{d}{dx}a^x = a^x \ln a$; chain rule brings down k.	Differentiating a number-to-a-power model.	Factor is $k \ln a$, never $\ln(kx)$.	[T][S]
(c) $\dfrac{dV}{dt} = 100 \cdot 10^{0.05t} \cdot 0.05 \ln 10 = 0.05 \ln 10\, V$	Differentiate the **exact** base 10.	Applying the rule to V.	Using rounded 1.12 gives $T = 13.10$ (still rounds to 13 — luck, not method).	[T]
(c) Solve $= 50$: $10^{0.05T} = \dfrac{10}{\ln 10}$, $T = \dfrac{\log_{10}(10/\ln 10)}{0.05}$	Show algebra (calculator-only barred).	Setting derivative $= 50$.	Must show working — "solutions relying entirely on calculator technology are not acceptable."	[T][Q]
(c) $T = 12.755\ldots \Rightarrow \boxed{T = 13}$	Round only the final T to nearest integer.	Final answer.	Round at the end, to the stated precision.	[T][Q]

[T] Drill set Harry assigned (do closed-book until automatic): $\frac{d}{dx}3^x = 3^x \ln 3$; $\frac{d}{dx}3^{6x} = 3^{6x} \cdot 6 \ln 3$ (**not** $\ln(6x)$); $\frac{d}{dx}5^{2x} = 5^{2x} \cdot 2 \ln 5$; $\frac{d}{dt}1.12^t = 1.12^t \ln 1.12$. This is the **next-session starter** Harry said he would test.

---

5. R-form, trig equation & optimisation table (Q4)

From the paper [Q]: (a) express $f(x) = \sqrt{3}\sin 2x - 3\cos 2x$ as $R\sin(2x - \alpha)$, $R > 0$, $0 < \alpha < \frac{\pi}{2}$, **exact** R, α. (b) for $g(x) = \dfrac{18}{f(3x) + 4\sqrt{3}}$ $(x > 0)$: (i) exact minimum of g; (ii) smallest x where it occurs. *"Show all stages of working."* **[6 marks]**

Method / line	Conditions & meaning	Trigger	Trap	Src
Expand $R\sin(2x - \alpha) = (R\cos\alpha)\sin 2x - (R\sin\alpha)\cos 2x$	Compound-angle identity from the formula book (minus version).	Any "$a \sin - b \cos$ into $R \sin$".	Pick the correct $\pm$ form; look it up, don't memorise.	[T][S]
Match: $R\cos\alpha = \sqrt{3}$, $R\sin\alpha = 3$	Equate $\sin 2x$ and $\cos 2x$ coefficients.	After expanding.	Line up which coefficient is which (a vs b).	[T][Q]
α**:** $\tan\alpha = \dfrac{3}{\sqrt{3}} = \sqrt{3} \Rightarrow \alpha = \frac{\pi}{3}$	Divide the two equations; use $\tan^{-1}$.	Finding α exactly.	Radians; use $\tan^{-1}$ **not** $\sin^{-1}$.	[T]
R**:** square-and-add: $(\sqrt{3})^2 + 3^2 = 12 \Rightarrow R = 2\sqrt{3}$	$R^2(\cos^2\alpha + \sin^2\alpha) = a^2 + b^2$.	"Exact value of R".	Don't sub a decimal α back — "could be quite dangerous."	[T]
$\boxed{f(x) = 2\sqrt{3}\sin(2x - \frac{\pi}{3})}$	Assembled exact answer.	Answer to (a).	Keep exact; no decimals for an "exact" command.	[T][Q]
(b) $f(3x) = 2\sqrt{3}\sin(6x - \frac{\pi}{3})$	Replace $x \to 3x$ inside f (coefficient $2 \cdot 3 = 6$).	Composite $f(3x)$.	It's the *equation* that gets $x \to 3x$, not the coordinates.	[T]
(b-i) g min when denominator max: $\sin(6x - \frac{\pi}{3}) = 1$	Fixed positive numerator over a wave $\Rightarrow$ min at max denominator.	"Exact minimum" of a $c/D(x)$.	No calculus needed; reason from $\sin_{\max} = 1$.	[T]

Method / line	Conditions & meaning	Trigger	Trap	Src
(b-i) min $g = \frac{18}{6\sqrt{3}} = \frac{3}{\sqrt{3}} = \boxed{\sqrt{3}}$	Rationalise $\frac{3}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$.	Simplifying the min.	Rationalise the surd denominator.	[T][Q]
(b-ii) $6x - \frac{\pi}{3} = \frac{\pi}{2} \Rightarrow x = \frac{5\pi}{36}$	Principal solution of sin $= 1$; check smallest positive.	"Smallest x" for the minimum.	Verify it's smallest (next is $\frac{17\pi}{36}$); check $\pi - \theta$ partner.	[T][Q]

> **[T] Harry's exam warning:** for an *exact-value* question, **square-and-add** for R and **divide** for tan α — this always gives exact values; substituting a rounded α "could be quite dangerous."

---

6. Quotient-rule model maximum table (Q5)

From the paper [Q]: $N = \dfrac{4000e^{0.1t}}{19 + e^{0.2t}}$, $t \geq 0$. (a) squirrels at start; (b) $\dfrac{dN}{dt}$; (c) at max $t = T$, show $e^{0.2T} = A$; (d) max squirrels (nearest whole number). *"Show all stages of working."* **[8 marks]**

Method / line	Conditions & meaning	Trigger	Trap	Src
(a) $N = \frac{4000e^0}{19 + e^0} = \frac{4000}{20} = \boxed{200}$	$e^0 = 1$; "at the start" means $t = 0$.	"Number at the start."	Keep the "$+1$": denominator $= 19 + 1 = 20$, "be careful not to lose the one."	[T][Q]
(b) Quotient rule, $u = 4000e^{0.1t}$, $v = 19 + e^{0.2t}$	$\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$ (formula book).	Differentiating top-over-bottom.	Use quotient rule, not product-rule on uv^{-1}.	[T][S]
(b) $u' = 400e^{0.1t}$, $v' = 0.2e^{0.2t}$	$0.1, 0.2$ come down (chain rule on e^{kt}).	Differentiating each part.	Bring the coefficient down correctly.	[T]
(b) $\frac{dN}{dt} = \frac{7600e^{0.1t} - 400e^{0.3t}}{(19 + e^{0.2t})^2}$	$e^{0.1t}e^{0.2t} = e^{0.3t}$; $400 \cdot 19 - 800 = 7600 - 400$.	Assemble & simplify.	**Keep the minus sign** in the numerator.	[T][Q]
(c) Max $\Rightarrow \frac{dN}{dt} = 0$; denom $\neq 0$	Only the numerator can be zero; $(19 + e^{0.2T})^2 > 0$.	"At a maximum when $t = T$."	Don't set the denominator to zero.	[T]
(c) $e^{0.1T}(7600 - 400e^{0.2T}) = 0$; $e^{0.1T} > 0$	Factor out $e^{0.1T}$; it's always positive, so reject it.	Solving numerator $= 0$.	$e^x > 0$ always — reject $e^{0.1T} = 0$.	[T]
(c) $e^{0.2T} = \frac{7600}{400} = 19 \Rightarrow \boxed{A = 19}$	"Show that $e^{0.2T} = A$."	Answer to (c).	**Stop here** — it's "show that", don't solve for T yet.	[T][Q]
(d) $0.2T = \ln 19 \Rightarrow T = \frac{\ln 19}{0.2}$ (stored)	Base is e, so use ln, not $\log_{10}$.	"Hence find the maximum."	Keep T stored/unrounded.	[T][Q]
(d) $e^{0.1T} = \sqrt{19}$, denom $= 19 + 19 = 38$	$e^{0.1T} = (e^{0.2T})^{1/2} = \sqrt{19}$.	Substituting back.	Show the substitution — working is marked.	[T]

Method / line	Conditions & meaning	Trigger	Trap	Src
(d) $N = \dfrac{4000\sqrt{19}}{38} = 458.83\ldots \Rightarrow \boxed{459}$	Round only the final answer to nearest whole.	Final answer.	Use stored T; **459**, not the ledger's mis-recorded "959" (see §7).	[T][Q]

[T] Rounding technique Harry drilled: "we only round at the very final answer… use the stored value" of T — never a rounded T.

---

7. Tutor / exam technique & status-boundary table

Two things sit in this section: (i) the **conflicts** in the evidence pack and how they resolve; (ii) the **status boundaries** (what was solved, prompted, deferred or only discussed). Nothing here is invented; each row cites its tier.

7.1 Evidence conflicts (resolved)

Item	Conflict	Resolution & why	Src
ESAT option labels	Board's multiple-choice list transcribed as garbled ("$R = 5r$, $R = 10r$...") and does **not** contain $r(10+\sqrt{105})$.	Board options **untrustworthy**; rely on the on-screen derivation only, match the value to the option letter on the day.	[T]
Q4(a) coefficients	One video segment reads $3\sin 2x - 4\cos 2x$ ($R = 5$, $\alpha = 0.927$); other segment + paper read $\sqrt{3}\sin 2x - 3\cos 2x$.	**Paper is authoritative** and demands *exact* $\alpha = \frac{\pi}{3}$ (incompatible with 0.927). "$R = 5$" is a transcription error, discarded.	[Q] over [T]
Q5 denominator	One segment mis-copies $19 + e^{0.1t}$.	Paper + segment 4 give $19 + e^{0.2t}$; used throughout.	[Q]
Q5(d) "459 vs 959"	One segment states max $= 958.81 \to 959$.	Model gives $4000\sqrt{19}/38 = 458.83 \to 459$; transcript: Harry "459 is fine." **459** used; "959" is a flagged analysis error.	[Q][T]
Question numbering	One ledger header labels the IVT/cube-root question "Question 1".	Loom chapters, audio and paper all place it at **Q2** (real Q1 is f, g). Paper numbering followed.	[Q][T]

7.2 Status boundaries (what was actually done)

Item	Status	Evidence	Src
ESAT warm-up	**Solved** on board: $R = r(10+\sqrt{105})$; option letters not captured.	Board derivation.	[T]

Item	Status	Evidence	Src
Q1 (f, g)	**Discussed only.** Phuc said he'd done part (a); chose to move on. **No tutor working exists — none reconstructed.**	Transcript.	[T]
Q2(a)	**Reviewed** (previously attempted): sign change + continuity emphasised.	Transcript.	[T]
Q2(b)	**Completed with prompting — transcript/ledger status conflict.** Transcript confirms target reached; video ledger says partial.	Transcript vs ledger.	[T]
Q2(c)	**Skipped / deferred** — "we'll leave that one"; never started.	Transcript.	[T]
Q3(a)(b)(c)	**Solved:** $0.05t + 2$; $a = 100, b = 10^{0.05}$; $T = 13$.	Transcript + paper.	[T][Q]
Q4(a)(b)	**Solved:** $R = 2\sqrt{3}, \alpha = \frac{\pi}{3}$; min $g = \sqrt{3}$; $x = \frac{5\pi}{36}$.	Transcript + paper.	[T][Q]
Q5(a)–(d)	**Solved:** 200; $\frac{dN}{dt}$; $A = 19$; **459**.	Transcript + paper.	[T][Q]
Q6–Q9	**Not discussed** — out of scope this session.	Paper.	[Q]
Homework	**Study log:** differentiation revision, ESAT formulae, P3 June 2024 paper. **Transcript (separate):** finish Q2(c) and send to Harry. These are distinct sources — the study log does **not** list "finish Q2(b)".	Study log + transcript.	[T]

7.3 Exam-technique boundaries Harry named

Technique	Statement	Src
Command words	"Show that" = reach the printed line and stop (Q5c); "hence" = use the previous part (Q5d).	[T][Q]
Exact vs rounded	Exact = surds/fractions/π, no decimals; square-and-add for exact R.	[T]
Rounding	Round only the final answer; substitute the *stored* value.	[T]
Formula book	Look up the compound-angle identity and quotient rule rather than memorising; radians, $\tan^{-1}$.	[T][S]
Calculator-only bar	Q1 and Q3 print "solutions relying entirely on calculator technology are not acceptable" — show algebra.	[Q]
October entry plan	A* in every module by October is "a real stretch"; may be "safer to push some back" and enter only the **3–4** modules Phuc is most ready for. Find out centre deadlines.	[T]

8. Sources and status boundaries (compact)

- **Primary — [T] Tutor:** Loom recording (1 h 08 m 23 s), `https://www.loom.com/share/bd86ba51060a46aba65384b65b59338f`, via the spoken transcript `sources/loom_transcript.txt` (primary) and the four-segment video ledger `analysis/gemini_video_evidence.md` (secondary; contains the OCR/arithmetic errors reconciled in §7). Tutor **Harry** (Get Me That Grade), student **Phuc**, 11 July 2026.
- **Authoritative — [Q] Paper:** Pearson Edexcel IAL Pure Mathematics **P3, WMA13/01, Tue 21 Oct 2025** (`sources/WMA13_Oct2025_QP.txt`). Wins on any numerical/wording conflict with the video ledger.
- **Reference — [S] Spec/formula book:** Edexcel IAL Mathematics 2018 specification (`shared/Edexcel_IAL_Mathematics_2018_specification.txt`): 6.1 location of roots (sign change + continuity), 6.2 iteration, 2.3 R sin form, 3.3 log-linear graphs, 4.1–4.4 differentiation and $\frac{d}{dx}a^x = a^x \ln a$. Used only for clearly-labelled formal expansions; never voiced as Harry.
- **Cross-check only:** SaveMyExams topic pages — consulted as method sanity checks, never allowed to overwrite tutor/paper/spec evidence.

Status rules (load-bearing): Q2(b) = *completed with prompting (status conflict)*; Q2(c) = *skipped/deferred*; Q1 = *discussed only, no working*; ESAT option letters = *unresolved*; Q4(a)/Q5-denominator/“459” = *paper-authoritative, ledger errors flagged*. This pack does not export to Notion, publish, or claim downstream release — the parent verifies and exports.